

Musical Therapy For Stress Reduction Using Machine Learning And IoT

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Abstract: Living with stress has become a part of our lifestyle. Long-term health issues are closely related to stress. Now a days peoples are no exceptions; they undergo stress and strains in facing new situations, where the outcome is uncertain. They are unable to identify the sources of their stress because of their numerous concerns. Because a doctor cannot always monitor stress levels and diagnose it, stress is strongly linked to many symptoms of anxiety, including restlessness and fatigue, as well as symptoms of depression, such irritability and problems concentrating. Stress is a prevalent problem in modern society, and there is a growing interest in using technology to help people manage their stress levels. In order to identify the right raga to assist reduce stress, the proposed study investigates the use of IoT sensors to assess physiological signs linked to stress. Musical therapy has a long history in India's holistic medicine, an ancient medical discipline. People everywhere are paying attention to music medical therapy. A wearable device with built-in IoT sensors is developed to collect data by continuously monitoring heart rate, blood pressure, and an AI model that deals with calculating stress level is built based on this data. Different Ragas and Music are suggested based on the stress level in order to successfully deal with stress and provide emotional comfort. The study demonstrates the potential of IoT sensors and machine learning algorithms to develop personalized stress management systems that can help individuals cope with stress more effectively. The findings can be applied to various settings, such as workplaces and hospitals to enhance the well- being of individuals.

Keywords: Stress reduction, Music therapy, machine learning, IoT, Wearable device

I. INTRODUCTION

While stress has an adverse effect on health, relaxation promotes health. The current circumstances Stress is rising at the fastest rate. This explains why, despite riches, people are not happy. Stress is a state of pressure. There are three types of pressure: mental, physical, and emotional. When someone is under pressure, they believe they won't succeed. Stress is found in all places. It is found in every single personality. Stress can be brought on by any idea or circumstance. Nature, however, is distinct. There are several types of stress. It is economical, psychological, and bodily. Stress can be brought on by any idea or circumstance. Stress causes one to steadily rise in multiple denials.

Stressful circumstances completely alter a person's personality. Stress is also brought on by inferior and superiority complexes. It is rare for a disturbed man to perform anything well.

Stress is frequently harmful but can also be beneficial. You work harder and stay more engaged and motivated when you

experience constructive stress. Conversely, toxic stress leaves you feeling alone, fearful, and uninterested. As of right now, stress can be classified as either short-term or long-term, or chronic. Short-term stress is situational; it ends or disappears as soon as the circumstances change. Chronic stress is a persistent issue. It is risky as a result. Chronic stress can occasionally be primarily inherited or genetically based. It is capable of ending anyone's life. One dies from excessive stress. The first step to living a happy, stress-free life is accepting stress. It is critical to identify and treat stress before it becomes severe since there are inexpensive, simple solutions accessible.

In the modern world of rivalry and competition, stress has become a necessary component of life for all people and has a wide range of direct and indirect effects. Medical research has shown that stress aggravates dangerous illnesses that have the potential to be fatal.

Stress is a growing problem that also has an impact on each person's physical and mental health and personal healthcare area. Music therapy is a useful tool for treating stress in order to help patients feel calm and at ease. This paper aims to propose a Musical Therapy system for Stress Reduction using machine learning, IoT. This system (android app) will play the raga when there is increase in stress level base on the heart rate and respective motions. This work tends to present development of an IOT-based wearable device that can collect real-time data from various sensors, such as heart rate, X-Y-Z axis to monitor the stress level of a person, and training of a machine learning model to accurately predict the stress level of a person based on the collected sensor data. This work also explains the design and development of an android application that can display the real- time stress level of a person, and development of a system that can suggest suitable music for stress reduction based on the predicted stress level. This work ensures the establishing seamless communication between the IOT devices and the android application to ensure real-time data transfer and monitoring. The propose system helps taking the time to relax every day helps manage stress and protect your body from the effects of stress.

II. LITERATURE SURVEY

In [1], research proposed a classification algorithm for stress prediction called Adaptive Neuro- Fuzzy Inference System assisted Fire Works Grey Wolf Optimization (ANFIS-FWGW). A sensor-integrated keyboard has been used in this machine- learning framework to estimate the stress levels of computer users. For feature se- lection, the LASSO algorithm is utilized. The ANFIS-FWGW classification algorithm exceeds the existing algorithms with an accuracy of 94 In paper

[2], researchers have examined physiological signs that accurately predict stress. These signals are gathered from different kinds of sensors. They are working on two datasets, the first of which was compiled using IOT sensors connected to each participant's chest. A second dataset was gathered using sensors in wearable technology. Deep Neural Networks were created to classify emotions and detect stress. CNN rates a 3-class classification task for emotion classification at 99.55

An in-depth analysis of the Raga's of Indian classical music (ICM) is provided in this research work, along with information on how they relate to Vaatha, Pitta, and Kapha, as well as the time of day when they are performed, the season of the year when they have their most beneficial effects, and other factors. The first case study pertains to the successful management of stress utilizing Darbari Kanada Raaga for patients having gastro endoscopy, while the second case study emphasizes the lowering of blood pressure and heart rate for pregnant hypertensive women in the third trimester. Both of these examples use melodic scales [3].

The [4] study uses an m-health solution and wearable physiological sensors to analyze the effects of a particular training protocol on the cognitive performances and stress response of a sample of frail older persons with mild cognitive impairment (MCI). They examine how the stress response of elderly and weak patients changes during therapy and how physical activity benefits cognitive training. Finally, based on a number of machine learning techniques, they assessed a stress detection system. The feature selection methods RF-Gain Ratio and RF-PCA were utilized. Using the Leave One Subject Out (LOSO) validation scheme, classification performance on one (or more) brand-new, unexplored subjects is assessed.

In paper [5], stress-related health data from the KNHANES VI dataset were analyzed. The most essential nine features from the dataset are chosen using statistical analysis techniques like the t- test and chi-square test. To test the viability of stress classification, the suggested model was assessed using Deep Belief Network (DBN) and on conventional machine learning techniques in addition to this clustering. Comparing classification abilities involved using a confusion matrix. The proposed model, which was integrated into a stress monitoring system, classified stress with an accuracy of 75.32.

In [6], they discussed the effects of Indian classical music on the body in their research. Every raga has a distinct characteristic of how people are treated. Each Raga is associated with a particular emotion or attitude that nature inspires in people. This work explained that there are several benefits of music therapy for the body. Music therapy is used to reduce stress and tension as well as to inspire recovery.

The science of extracting information from music is known as music information retrieval (MIR). They evaluate two computational models that accept two music signals as input and quantify how similar or dissimilar the signals are in order to produce a numerical measure that, in theory, correlates with human judgment. Learning raga motifs and developing the capacity to recall and perform the phrases in the context of performance are key components of IAM's pedagogy. A method where the two basic computing units are the distance between the two input signals and a representation produced from an auditory signal (one input vs. one reference signal). The

system's final result is the similarity rating expressed as a score [7].

The above survey shows that the existing systems are trained on open source dataset, additionally the real time result hasn't been mentioned. These systems use AI technology for prediction of stress but never recommend solution if the user has high stress. These systems cannot forecast real-time stress levels and also, these systems don't recommend any exercise or solution to reduce the stress.

According to the World Health Organisation (WHO), stress may be a psychiatric condition that shortens one in four voters' lives. Human stress causes mental health problems in addition to socioeconomic problems, opaque work environments, poor working relationships, depression, and in extreme situations, suicide. Stress is a growing problem that also has an impact on each person's physical and mental health and personal healthcare area. People of all ages and socioeconomic backgrounds are affected by stress, which has become a major problem in today's society. Stress makes muscles tense and impairs regular physical activity. A person's susceptibility to complex mental illnesses like borderline personality disorder (BPD), which can lead to harmful mood swings, altered behavioral patterns, and eating disorders, as well as stress levels that disturb our socioeconomic system, might increase with stress.

There are several ways to deal with stress, including exercise, meditation, and medicine. Musical therapy is one of the cutting-edge methods that uses music to promote relaxation and lower stress levels. The usage of AI, ML, and IoT can now be used to improve musical therapy because of the advancement of technology. The most popular music in music therapy is Indian raga.

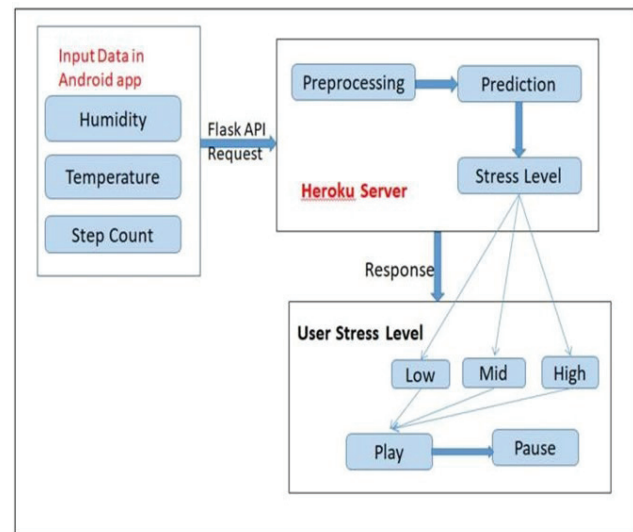


Fig. 1. Architecture diagram of System

In the field of medicine, music therapy is receiving increased attention, particularly in the treatment of patients with mental instability. To demonstrate that music therapy can speed up healing time, it can be used to replace medications now used to treat autism, and reduce stress, and other conditions.

To reduce stress, this research focuses on investigating the possibility of using AI, ML, and IoT in musical therapy. In this proposed system, a hand band with IOT sensors calculates

real-time data, pass it to an AI-ML model, and creates a system that can forecast the raga based on the user's stress level. Research is being done on the effects of music on stress reduction, the creation of an Android platform for delivering musical therapy, and the development of an AI and ML model for music selection.

It has been demonstrated that using Indian ragas in music therapy helps people feel less stressed and anxious. Stress is a common cause of risk for many different illnesses. It's evident that medical professionals are focusing more on music therapy, particularly when it comes to treating mentally unstable patients. Indian ragas are the key music used in music therapy. Therefore, this system is designed to incorporate Indian ragas in the suggested music for stress reduction.

III. PERFORMANCE REQUIREMENTS

Response Time: The software program has a fast response time, It is able to collect and process data in real-time, and suggest suitable music for stress reduction within seconds of detecting the stress level.

Memory Usage: The software program is optimized for memory usage and should not consume excessive memory, which could slow down the system or cause other issues.

Maximum Error Rate: The software program has a low error rate, meaning it is accurately predict stress levels and suggest suitable music for stress reduction. The maximum acceptable error rate is specified in the requirements.

Scalability: The software program is able to handle large amounts of data and support multiple users simultaneously. It is scalable to meet the needs of future growth.

Security: The software program is secure, meaning it should protect user data and prevent unauthorized access. It is able to handle potential security threats, such as data breaches or cyber- attacks.

IV. PROPOSED SYSTEM ARCHITECTURE

The implementation is divided into two development phases:

Phase 1: Android Application for Stress detection and Music recommendation. A mobile app is developed for model working and music recommendation during this phase.

Phase 2: Machine learning model is developed to estimate the stress level as low, mid, and high based on humidity, temperature, and step count.

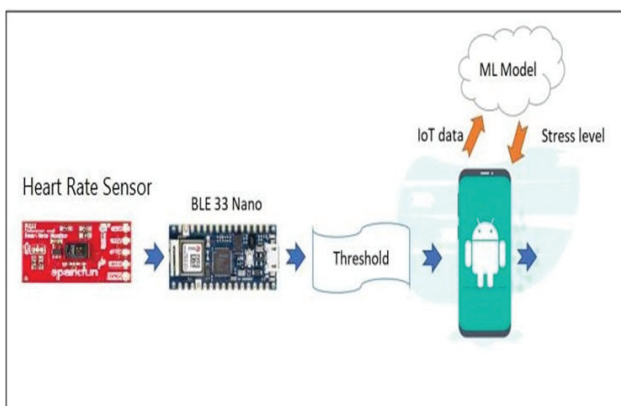


Fig. 2. Architecture diagram of training the APP

We have employed a variety of supervised classification techniques for the categorization of stress level, including Logistic Regression, Support Vector Machine, Decision Tree, Random Forest Classifier, and Ada Boost Classifier. Decision Tree, an algorithm, provided the best accuracy of 98 percent. As a result, we chose the Decision Tree Algorithm to forecast stress levels.

We have collected real time BPM Data from 29 different users. Based on this collected we have built the Machine Learning Clustering Algorithm for stress level prediction. This Machine learning model is deployed into android app for stress level prediction and Raga recommendation. Three different types of raga has been selected with respect to stress level based on their characteristics of Raga.

To establish a BLE connection between the Arduino Nano 33 BLE and an Android device, we have use the Bluetooth Gatt services in the Android SDK for the communication purpose. The Arduino Nano 33 BLE is a small and powerful development board based on the nRF52840 microcontroller. It is designed for wireless communication using Bluetooth Low Energy (BLE) and other protocols. The board has a 32-bit ARM Cortex-M4F core, 1MB flash memory, 256KB RAM, and a 9-axis accelerometer and gyroscope, temperature and humidity sensor, microphone, and an on-board antenna. It also has a wide range of communication interfaces, including I2C, SPI, UART, and USB which helps in data transmission between one devices to another with low latency.

The Arduino Nano 33 BLE board features a built-in Bluetooth Low Energy module that enables wireless communication between the board and other BLE- enabled devices, such as smartphones, tablets, and other IoT devices. The board uses the nRF52840 chip from Nordic Semiconductor, which is a highly efficient and low-power wireless chip that supports Bluetooth Low Energy and other wireless protocols for communication.

The SparkFun Pulse Oximeter and Heart Rate Sensor is a compact and low-cost sensor module that can measure both heart rate and blood oxygen saturation (SpO2) levels in the blood. It uses a light-based measurement technique known as photo- plethysmography (PPG) to detect changes in blood volume in the fingertip or ear- lobe.

The sensor module includes an infrared LED and a red LED that are used to shine light through the skin and into the blood vessels. The amount of light that is absorbed or reflected by the blood vessels varies with the pulsatile blood flow which is related to the heart rate and SpO2 levels. The sensor module also includes a photodetector that measures the amount of light that is transmitted or reflected back from the blood vessels.

The SparkFun Pulse Oximeter and Heart Rate Sensor is based on the Maxim Integrated MAX30102 sensor IC, which includes a high-sensitivity photodetector, an integrated analog front- end (AFE), and a digital signal processor (DSP). The sensor module also includes an on-board 3.3V regulator and level shifters for 3.3V and 5V compatibility.

The sensor module can be connected to a microcontroller or a single-board computer (SBC) using the I2C interface, and can be configured using the MAX30102 register map. The sensor module also includes an onboard voltage translator for compatibility with 5V microcontrollers.

V. RESULTS AND DISCUSSION

For stress level prediction various classification algorithms have been used such as Logistic regression, Support vector machine, Decision tree classifier, Ada boost classifier etc. The classification accuracy has been compared using confusion matrix. From the experiment, Decision Tree have the highest accuracy with 98% of accuracy. Model is successfully deploy into android application. Working of this phase is shown in figure 3. We have gathered real-time data from IOT sensors in the project's second phase, including heart rate, Different clustering algorithms, such as K-means clustering,

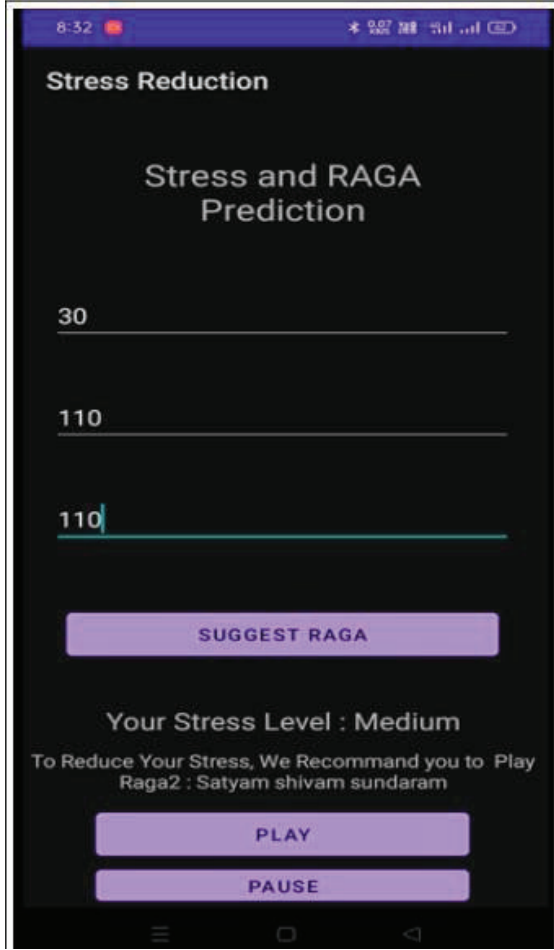


Fig. 3. Working of Android App

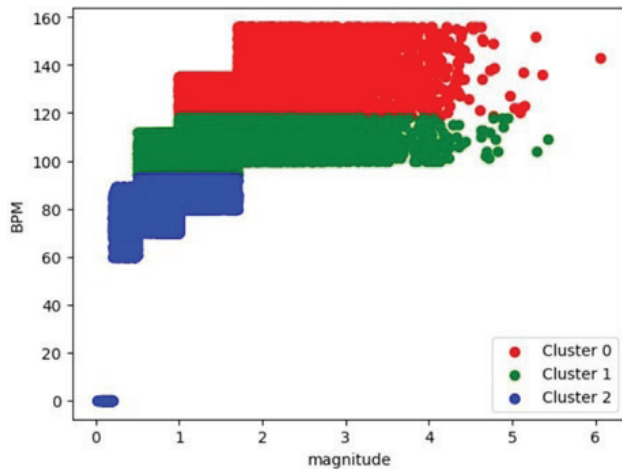


Fig. 4. Clusters form by K-means

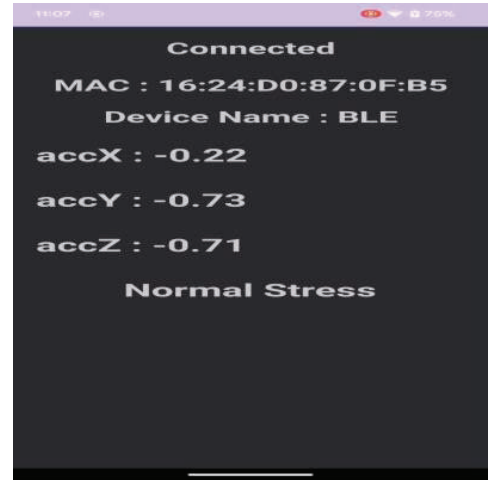


Fig. 5. working of App after stress detection

Hierarchical clustering, and Agglomerative Clustering, are trained based on the collected data in order to predict the stress level. All clustering algorithms are compared using the silhouette score. K-means has provided the finest silhouette score of 52% among the rest, as shown in table1.

TABLE I. COMPARISON OF SILHOUETTE SCORE FOR DIFFERENT CLUSTERING ALGORITHM

Sr. No.	Algorithms	Silhouette Score
1	K-Means	52
2	Agglomerative	46
3	Hierarchical	49

VI. CONCLUSION AND FUTURE SCOPE

In this paper, we have collected real-time data from sensors, based on this collected data we build the unsupervised K means clustering model for stress level prediction. This Machine Learning model is called by android application. Thus this android application is a user friendly platform for stress level calculation and Reduction by playing the Raga.

In mobile application there is scope for improvement of this App. App is not providing the SMS or Notification integration. Hence, it can be modified to give the SMS or Notification integration to user about their stress level. Apart from these there is scope for generating many more features. In the future we can add more sensors to add more parameters which contribute to measure the stress.

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